19CSE202
DATABASE MANAGEMENT SYSTEMS (DBMS)

Introduction To Relational Model

session 1

TOPICS

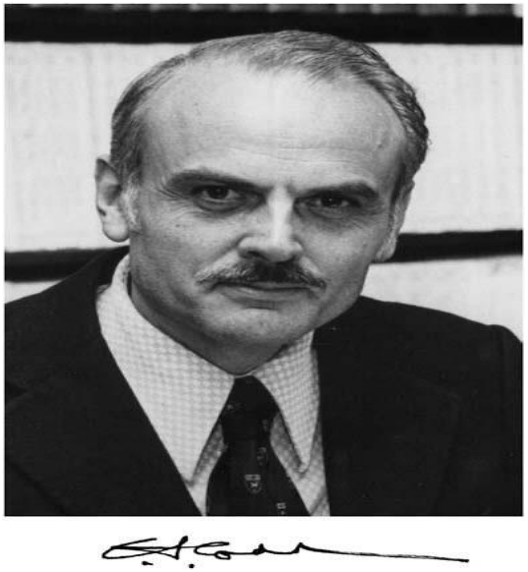
- Introduction to RDBMS & Relational Model
- Relations
- Schema
- Attributes
- Keys
- Referential Integrity
- Schema Diagram
- Relational Query Languages

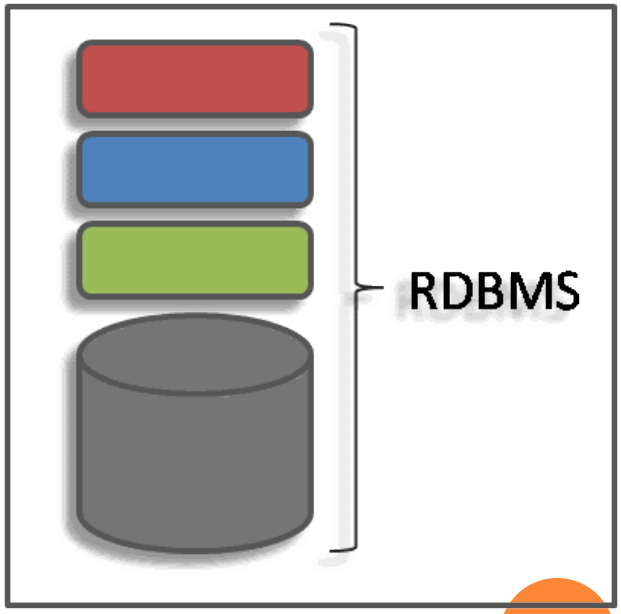


Relational Database Model



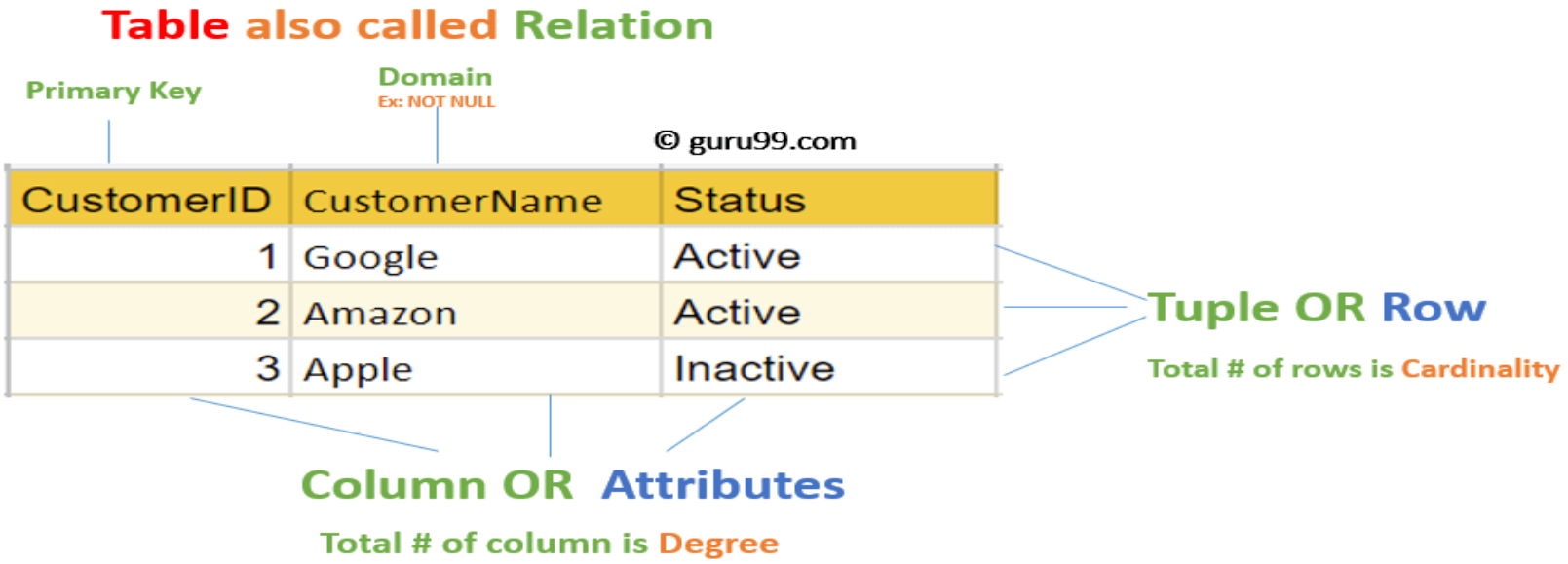
- Most Common model in industry
- Based on the relational model developed by E.F.Codd
- Data and relations between them are organized as tables





Relational Database Model

- Data is presented as a collection of **relations**.
- Each **relation** is depicted as a **table**.
- **Columns** are **attributes**.
- Each **row** represents an **entity.**



Relations

- A relation is a two-dimensional table:
  - Relation ≈ table.
  - Attribute ≈ column name.
  - Tuple ≈ row (not the header row).
- Database ≈ collection of relations.
- A relation has two parts:
  - Schema defines column heads of the table (attributes).
  - Instances contains the data rows (tuples, rows, or records) of the table.

Student_name	Course	Grade
Hermione Grainger	Potions	A-
Draco Malfoy	Potions	B
Harry Potter	Potions	A
Ron Weasley	Potions	C

SCHEMA

- **SCHEMA** describes the relation name (table name), attributes, and their names.

CoursesTaken :

Student_name	Course	Grade
Hermione Grainger	Potions	A-
Draco Malfoy	Potions	B
Harry Potter	Potions	A
Ron Weasley	Potions	C

CoursesTaken(Student, Course, Grade)

- A design in a relational model consists of a set of schemas.
- Such a set of schemas is called a **relational database schema**.



Degree and Cardinality

CoursesTaken :

	1 Student_name	2 Course	3 Grade
1	Hermione Grainger	Potions	A-
2	Draco Malfoy	Potions	B
3	Harry Potter	Potions	A
4	Ron Weasley	Potions	C

- **Degree** is the number of fields/attributes in schema (=3 in the table above)
- **Cardinality** is the number of tuples in relation (=4 in the table above)



INSTRUCTOR

ID	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000
12121	Wu	Finance	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
32343	El Said	History	60000
33456	Gold	Physics	87000
45565	Katz	Comp. Sci.	75000
58583	Califieri	History	62000
76543	Singh	Finance	80000
76766	Crick	Biology	72000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000

- Write down the Schema

INSTRUCTOR(ID,name,dept_name,salray)



INSTRUCTOR

ID	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000
12121	Wu	Finance	90000
15151	Mozart	Music	40000
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83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000

- What is the degree and cardinality

Degree= 4

Cardinality= 12



Attribute Types

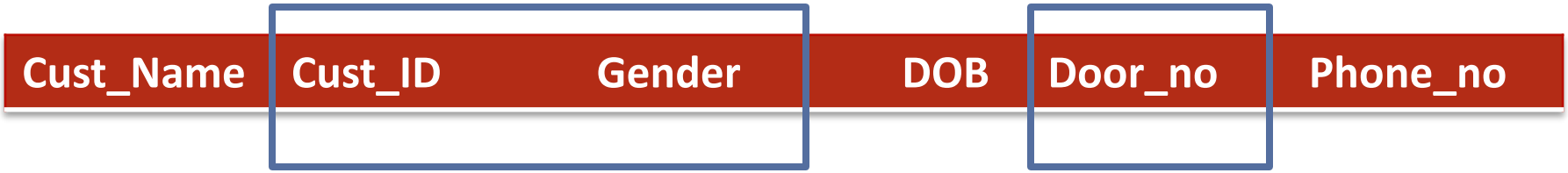
- The set of allowed values for each attribute is called the **domain** of the attribute
  - **Atomic Attribute**
  - **Composite Attribute**
  - **Single Valued Attribute**
  - **Multivalued Attribute**
  - **Stored Attribute**
  - **Derived Attribute**
  - **Null Valued Attribute**
  - **Key Attribute**



Attribute Types

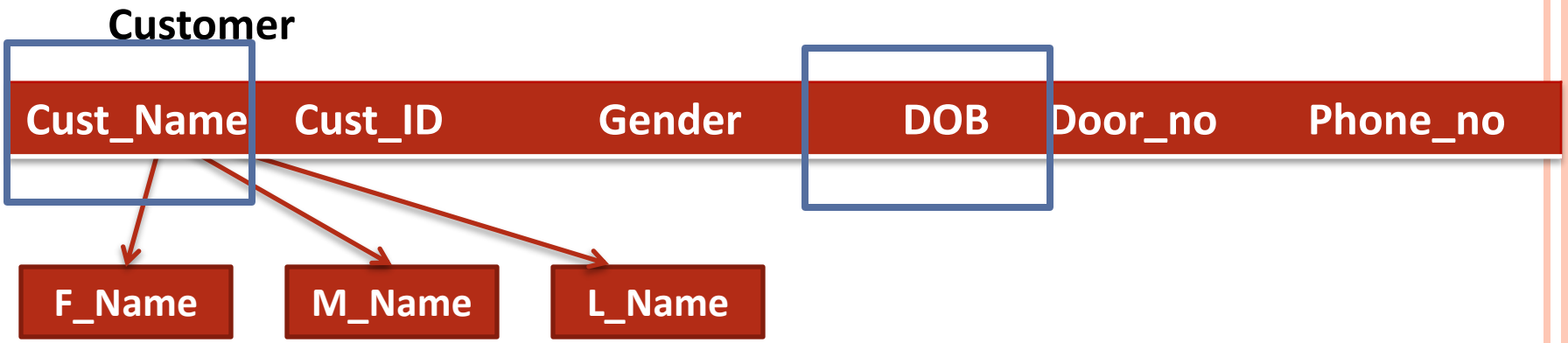
- An attribute that cannot be divided into smaller independent attribute is known as **atomic attribute**.

Customer



Attribute Types

- An attribute that can be divided into smaller independent attribute is known as **composite attribute**.



Attribute Types

- An attribute that has only single value for an entity is known as **single valued attribute**.

- An attribute that can have multiple values for an entity is known as **multi valued attribute**.

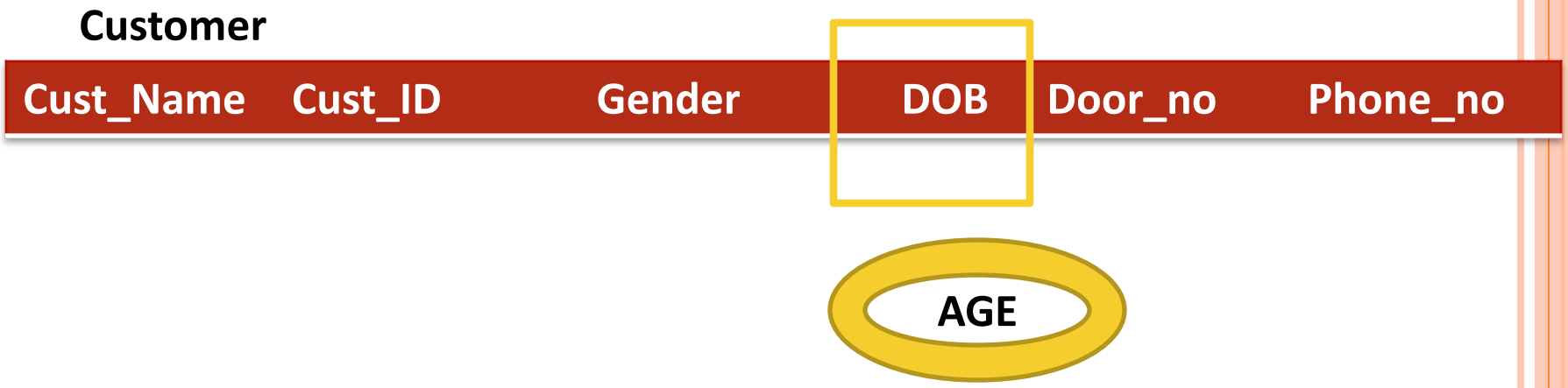
Customer

Cust_Name	Cust_ID	Gender	DOB	Door_no	Phone_no

Attribute Types

- An attribute that cannot be derived from another attribute is known as **stored attribute**.

- An attribute that can be derived from another attribute is known as **derived attribute**.



Attribute Types

- An attribute, which has not any value for an entity is known as **null valued attribute.**

- An attribute that has unique value of each entity is known as **key attribute.**

Customer

Cust_Name	Cust_ID	Gender	DOB	Door_no	Phone_no

Attribute Types

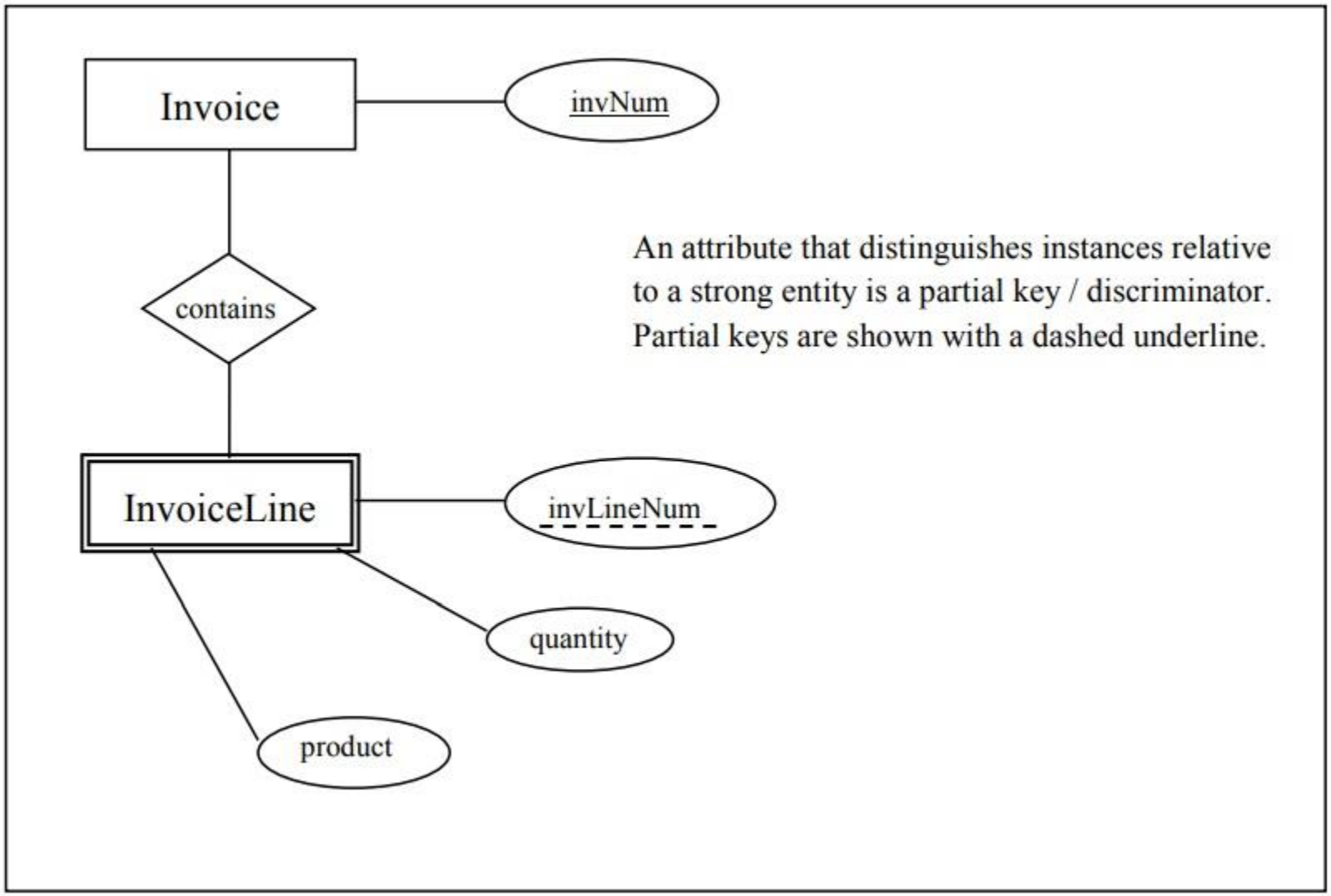
Student

ID	Student name	Course	Grade	Address	Phone_number
1	Hermione Grainger	Potions	A+	House no 5, Little Whinging, Surrey	9191819191 7877878778
2	Draco Malfoy	Potions	B	House no 10, Little Whinging, Surrey	2347879980
3	Harry Potter	Potions	A	House no 4, Privet Drive, Little Whinging, Surrey	1234567890
4	Ron Weasley	Potions	C	House no 12, Little Whinging, Surrey	5689745623

Identify **Atomic, Composite, single valued ,multi valued, null valued and key** attributes



Partial key/Discriminator Attribute



KEYS

- A data item [attribute] which can uniquely identify a tuple in a relation.
  - ✓ **SUPER KEY**
  - ✓ **CANDIDATE KEY**
  - ✓ **PRIMARY KEY**

SUPER KEY

- **Any combination of attributes which can uniquely identify a tuple**

Student

s_name	S_roll
ram	123
sanjay	124
manju	125
meenu	126
ram	127

Super Key={S_roll} , {s_name,S_roll}

Identify super keys for the following table

Book

book_id	name	author
B1	XYZ	A1
B2	ABC	A1
B3	XYZ	A2
B4	PQR	A3
B5	LSP	A1

SK={ book_id} , {book_id,name}, {name,author}, {book_id,name,author}

- **Super key will be redundant**



Candidate key

- A super key without redundancy
- Candidate key may have multiple attributes
- Not reducible
- Minimal super key

{ book_id}

{book_id,name}

{name,author}

{book_id,name,author}

Book

book_id	name	author
B1	XYZ	A1
B2	ABC	A1
B3	XYZ	A2
B4	PQR	A3
B5	LSP	A1



KEYS

Employee

Emp_Id	Emp_Number	Emp_Name
E01	2264	Steve
E22	2278	Ajeet
E23	2288	Chaitanya
E45	2290	Robert

How many super keys the above table can have?

1. {Emp_Id}
2. {Emp_Number}
3. {Emp_Id, Emp_Number}
4. {Emp_Id, Emp_Name}
5. {Emp_Id, Emp_Number, Emp_Name}
6. {Emp_Number, Emp_Name}



KEYS

Employee

Emp_Id	Emp_Number	Emp_Name
E01	2264	Steve
E22	2278	Ajeet
E23	2288	Chaitanya
E45	2290	Robert

How many candidate keys the above table can have?

1. {Emp_Id} ----No redundant attributes
2. {Emp_Number} ---- No redundant attributes
3. {Emp_Id, Emp_Number}--- Redundant attribute.
4. {Emp_Id, Emp_Name}--- Redundant attribute Emp_Name.
5. {Emp_Id, Emp_Number, Emp_Name} Redundant attributes. Emp_Id or Emp_Number alone are sufficient
6. {Emp_Number, Emp_Name} --- Redundant attribute Emp_Name.

KEYS

Employee

Emp_Id	Emp_Number	Emp_Name
E01	2264	Steve
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E23	2288	Chaitanya
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How many candidate keys the above table can have?

1. **{Emp_Id} ----No redundant attributes**
2. **{Emp_Number} ---- No redundant attributes**
3. {Emp_Id, Emp_Number}--- Redundant attribute.
4. {Emp_Id, Emp_Name}--- Redundant attribute Emp_Name.
5. {Emp_Id, Emp_Number, Emp_Name} Redundant attributes. Emp_Id or Emp_Number alone are sufficient
6. {Emp_Number, Emp_Name} --- Redundant attribute Emp_Name.

Primary key

- One of the candidate key

  { book_id}

  {name,author}

- Composite/Atomic

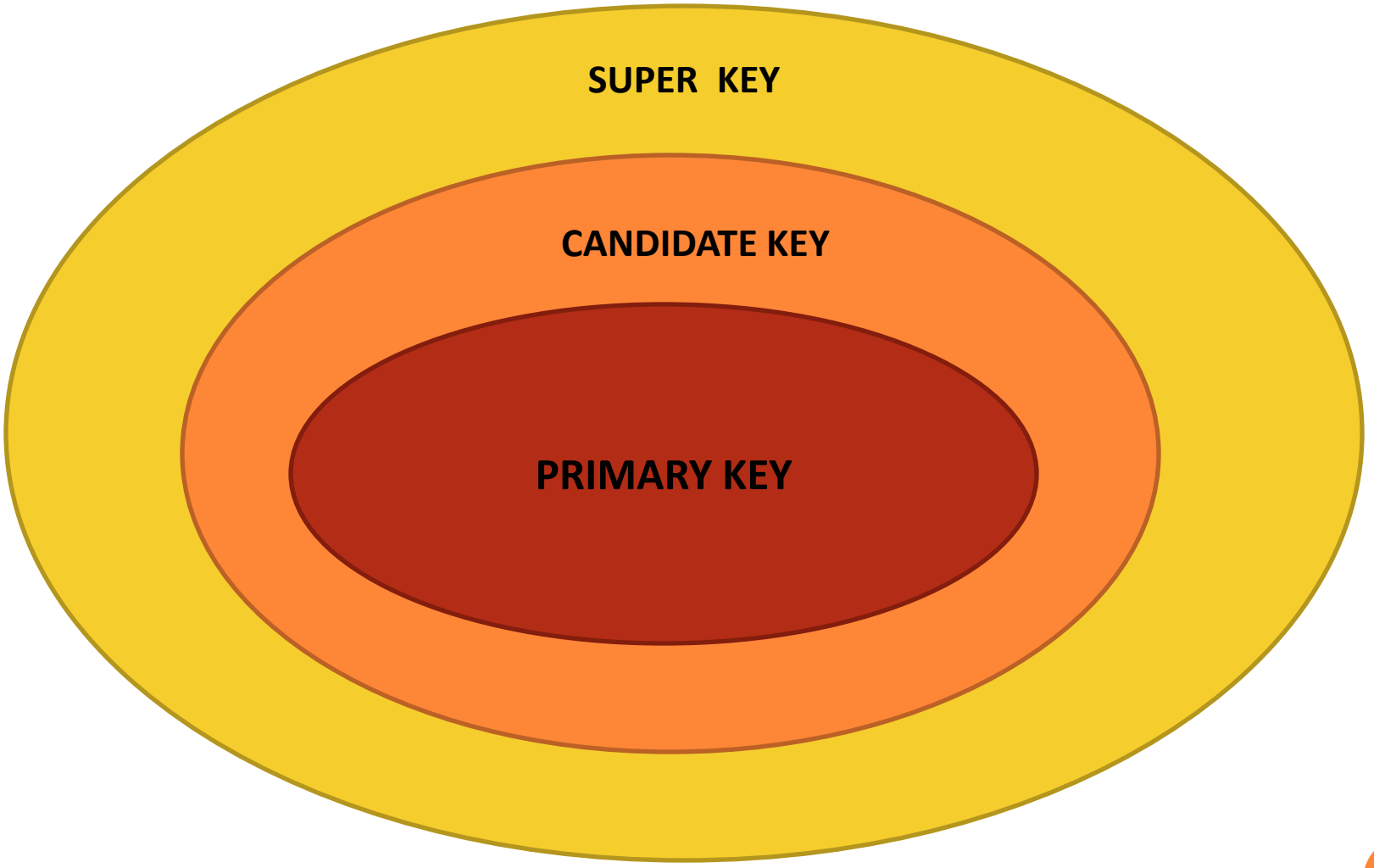
Book

book_id	name	author
B1	XYZ	A1
B2	ABC	A1
B3	XYZ	A2
B4	PQR	A3
B5	LSP	A1

How will you select a primary key ?



KEYS



Is everything fine here?

Department

DeptCode	DeptName
001	Science
002	English
005	Computer

Employee

Teacher ID	Fname	Lname
B002	David	Warner
B017	Sara	Joseph
B009	Mike	Brunton

HOW DO YOU GET THE DEAPARTMENT NAME OF A TEACHER (SARA)?

Employee

Teacher ID	Fname	Lname	DeptCode
B002	David	Warner	001
B017	Sara	Joseph	002
B009	Mike	Brunton	005



FOREIGN KEY:

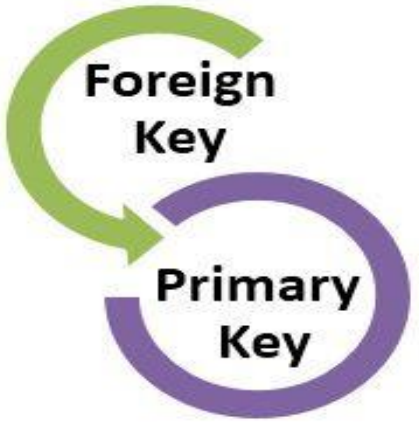
- Foreign keys are the attributes (columns) of a table that points to the primary key of another table.
- They act as a cross-reference between tables.
- r1,r2 ← relations

A relation r1 can have an attribute which will be the primary key of another relation r2

This attribute is known as foreign key from **r1 referencing r2**

r1 – referencing relation

r2- referenced relation



Foreign key

Department

DeptCode	DeptName
001	Science
002	English
005	Computer

Employee

Teacher ID	Fname	Lname
B002	David	Warner
B017	Sara	Joseph
B009	Mike	Brunton

Referenced Relation

Employee

Teacher ID	Fname	Lname	DeptCode
B002	David	Warner	001
B017	Sara	Joseph	002
B009	Mike	Brunton	005

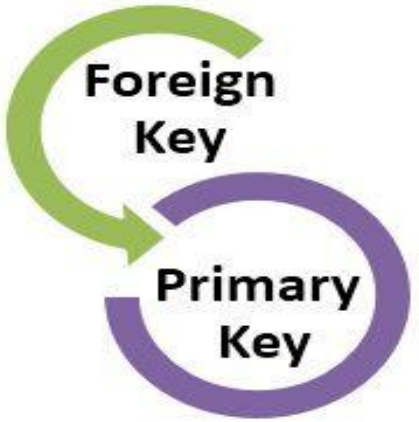
Referencing Relation



Referential Integrity

- Referential integrity refers to the accuracy and consistency of data within a **relationship.**
- In relationships, data is linked between two or more tables. (FK,PK)
- Need to ensure that data on both sides of the relationship remain intact.

So, referential integrity requires that, whenever a foreign key value is used it must reference a valid, existing primary key in the parent table.



Referential Integrity

Department

DeptCode	DeptName
001	Science
002	English
005	Computer

Delete

Referenced Relation

Employee

Teacher ID	Fname	Lname	DeptCode
B002	Chris	Evans	001
B017	Scarlet	Johansson	002
B009	Mark	Rufallo	005
B010	**Robert**	**Downy**	**008**

Referencing Relation



Referential Integrity

So referential integrity will prevent users from:

- Adding records to a related table if there is no associated record in the primary table.
- Changing values in a primary table that result in orphaned records in a related table.
- Deleting records from a primary table if there are matching related records.

Primary Key Constraint

- No column that is part of the primary key can contain a null.
- A table can have only one primary key

Foreign Key Constraint

constraint prevents any actions that would destroy link between tables with the corresponding data values

Integrity Constraints

- Integrity constraints are a set of rules. It is used to maintain the quality of information.
- Integrity constraints ensure that the data insertion, updating, and other processes have to be performed in such a way that data integrity is not affected.
- Used to guard against accidental damage to the database.

- ✓ **Domain constraints**
- ✓ **Entity integrity constraints**
- ✓ **Referential Integrity Constraints**
- ✓ **Key constraints**



Domain Constraints

- Domain constraints can be defined as the definition of a valid set of values for an attribute.
- The data type of domain includes string, character, integer, time, date, currency, etc. The value of the attribute must be available in the corresponding domain.

ID	NAME	SEMENSTER	AGE
1000	Tom	1st	17
1001	Johnson	2nd	24
1002	Leonardo	5th	21
1003	Kate	3rd	19
1004	Morgan	8th	A

Entity Integrity Constraints

- The entity integrity constraint states **that primary key value can't be null**.
- This is because the primary key value is used to identify individual rows in relation and if the primary key has a null value, then we can't identify those rows.
- A table can contain a null value other than the primary key field.

EMPLOYEE

EMP_ID	EMP_NAME	SALARY
123	Jack	30000
142	Harry	60000
164	John	20000
	Jackson	27000

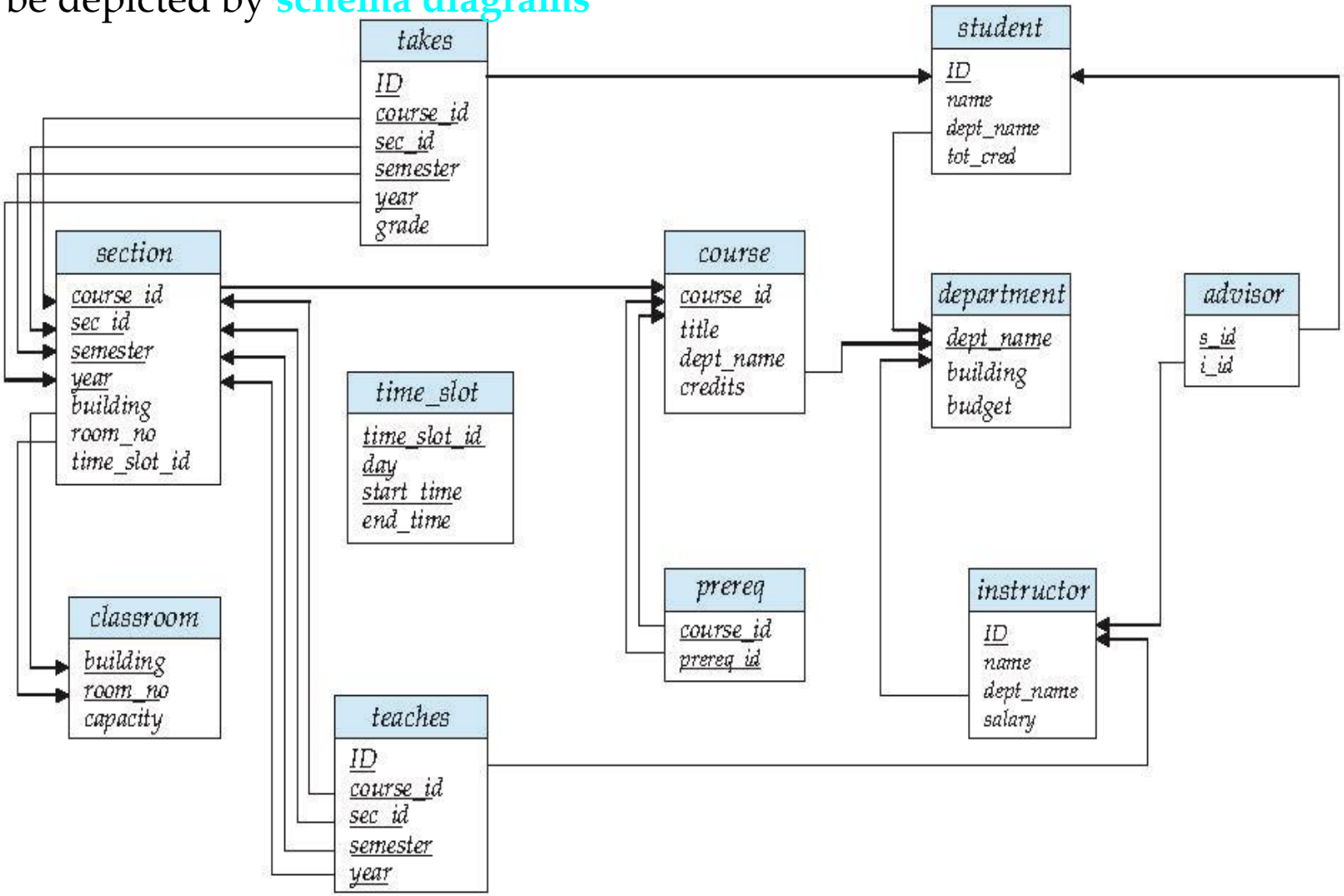
Key Constraints

- Keys are the entity set that is used to identify an entity within its entity set uniquely.
- An entity set can have multiple keys, but out of which one key will be the primary key. **A primary key can contain a unique value** in the relational table.

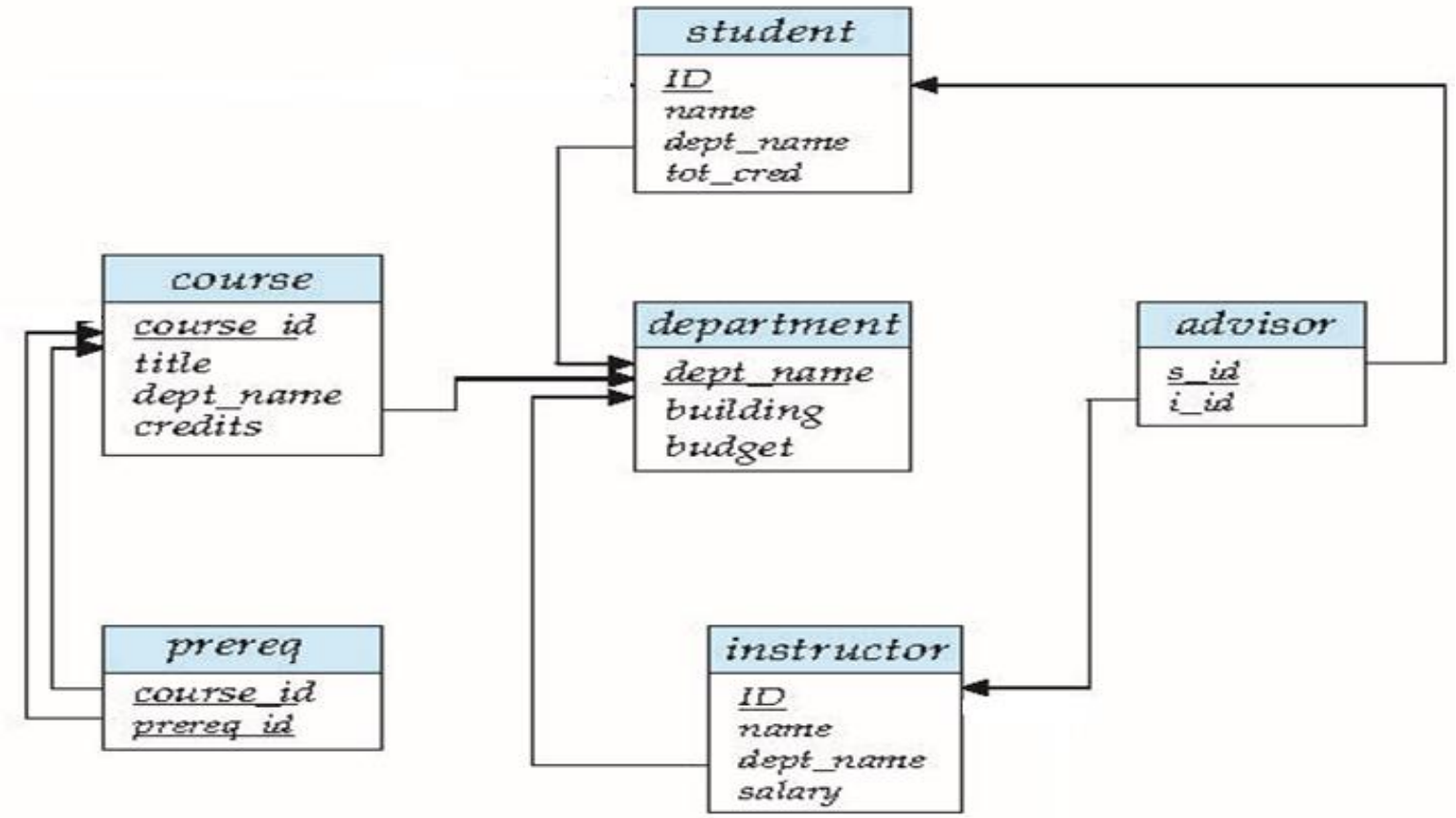
ID	NAME	SEMENSTER	AGE
1000	Tom	1st	17
1001	Johnson	2nd	24
1002	Leonardo	5th	21
1003	Kate	3rd	19
1002	Morgan	8th	22

Schema Diagram for University Database

A database schema, along with primary key and foreign key dependencies, can be depicted by **schema diagrams**



Schema Diagram for University Database



READ and REFER..

1.6.2 Database Design for a University Organization [Page NO: 16]
2.2 Database Schema [Page NO: 42- 47]

Constraints in SQL

- NOT NULL - Ensures that a column cannot have a NULL value
- UNIQUE - Ensures that all values in a column are different
- PRIMARY KEY - A combination of a NOT NULL and UNIQUE. Uniquely identifies each row in a table
- FOREIGN KEY - Prevents actions that would destroy links between tables
- CHECK - Ensures that the values in a column satisfies a specific condition
- DEFAULT - Sets a default value for a column if no value is specified



Summary

- Relational Model
- Schema
- Degree, Cardinality
- Attributes Types
- Keys
  - Super Key
  - Candidate Key
  - Primary Key
- Constraints

